NAME

mbpy_wcdview − Python water column viewer, an example of the mbio_py Python API.

VERSION

Version 5.0

SYNOPSIS

mbpy_wcdview
[
−−input=*file* {**−I***file*}
file
−−help {**−H**}
]

DESCRIPTION

mbpy_wcdview displays water column data, one swath at a time, read through the MBIO library with **mbio_py**, the Python API to MBIO installed with MB-System in *<prefix>/share/mbsystem/python*. It is written in Python with PySide6 (Qt6 for Python) and matplotlib, and serves as an example application of the API; **mbwcdview**(1) is the faster C++/Qt6 program for the same task.

Opening a Kongsberg *.kmwcd, *.kmall, *.wcd or *.all file finds the other file of the survey line in the same directory (same line number, nearest file name time) and matches the pings by ping number and time. The *.kmall or *.all supplies the soundings, navigation, heading and transducer depth; the *.kmwcd or *.wcd the water column. A *.all without a *.wcd supplies the water column too, if it holds any. Either may be missing: without a *.kmall or *.all the navigation comes from the position records of the *.kmwcd or *.wcd and depths are given below the transducer.

The window has an Open button, Save Image (twice the screen resolution, with the file name and ping information), check boxes **Bottom** (soundings of the matching survey ping: red amplitude and dark blue phase detections, filled good, hollow flagged), **WC bottom** (the bottom detections recorded in the water column), **Ray trace (SVP)** (samples traced through the logged sound velocity profile; on when a profile is found) and **Pick** (positions clicked in the image go into a table that can be saved as text), the depth and amplitude ranges, a fixed across track scale, a replay rate, and a slider that steps one swath at a time (also the mouse wheel and the left and right arrow keys). Amplitudes are in dB; the fill value of Kongsberg kmall data is drawn as no data. In *.wcd and *.all data the lowest value is a valid level and is drawn with the lowest color.

mbpy_wcdview needs Python >= 3.11, numpy, matplotlib and PySide6, and an MB-System libmbio with **mb_extract_wc**(3). The README in *<prefix>/share/mbsystem/python* explains the installation of these packages and the use of the API.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian dos Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

OPTIONS

−−input=*file* {**−I***file*}
The *.kmwcd, *.kmall, *.wcd or *.all to open; it may also be given as the last argument.

−−help {−H}
Print the usage message and exit.

ENVIRONMENT

MBSYSTEM_LIBMBIO
Full path of the libmbio to use instead of the one of the installation.

MBIO_PY_PATH
Directory containing mbio_py, instead of the installation's.

EXAMPLES

To view a Kongsberg survey line:
mbpy_wcdview 0080_20250618_223434.kmwcd

To run it with numpy, matplotlib and PySide6 from a virtual environment:
~/mbpy/bin/python $(which mbpy_wcdview) 0080_20250618_223434.kmall

SEE ALSO

mbsystem(1), **mbwcdview**(1), **mbio**(3)

BUGS

Stepping through the pings is slow compared with **mbwcdview**, as matplotlib redraws every cell of a swath. All water column pings of a file are kept in memory (about two bytes per sample).